

This manual is for installing ROS Melodic and SLAM Hector on Jetson Nano.

Prerequisites:

1. Properly flashed SD card with the most recent jetpack version
2. Ethernet or WiFi connection
3. Admin permission on the Jetson's Ubuntu GUI
4. Keyboard, mouse, monitor connected to the Jetson
5. RP LiDAR A2M8 connected to the jetson by USB-A

Step-1: Setup sources.list

Run:

```
sudo sh -c 'echo "deb http://packages.ros.org/ros/ubuntu $(lsb_release -sc) main" >
/etc/apt/sources.list.d/ros-latest.list'
```

Step-2: Set up keys

Run:

```
sudo apt install curl
curl -s https://raw.githubusercontent.com/ros/rosdistro/master/ros.asc | sudo apt-key add -
```

Step-3: Installation

Run:

```
sudo apt install ros-melodic-desktop-full
```

Step-4: Environment Setup

Run:

```
echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc
source ~/.bashrc
```

Step-5: Dependencies for building packages

Run:

```
sudo apt install python-rosdep python-rosinstall python-rosinstall-generator python-wstool
build-essential
```

Step-6: Initialize rosdep

```
sudo apt install python-rosdep
then
sudo rosdep init
then
rosdep update
```

Verification:

Run `rosversion -d`. It should return melodic.

Installing Hector SLAM on Jetson Nano

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6. ROS Melodic

Step-1:

Sudo apt-get update

Step-2:

sudo apt install ros-melodic-hector-slam